The regular polytopes of the third Conway group

Human Compression project: L071 recovery

Working manuscript; separate scientific review pending

We retain the complete proofs of enumeration by stabilizers of initial strings and of its implementation through double cosets in centralizers. Isomorphisms preserve the order of the distinguished generators. In particular, a polytope and its dual are counted separately unless they are isomorphic. This is the convention of Leemans–Mixer, *Algorithms for classifying regular polytopes with a fixed automorphism group*, Contributions to Discrete Mathematics 7(2) (2012), 105–118, Section 7.

Theorem 1. *The third Conway group is the automorphism group of exactly* 22 908 *regular polytopes up to isomorphism. Their rank distribution is*

rank	3	4	5
number	21 118	1 746	44.

Every other rank has count zero. The complete distribution by ordered Schläfli type agrees with every entry of the tables in Sections 7.1–7.3 of the source, including every printed zero and every unlisted type. The exact comparison data comprise 225 *printed rank-three cells,* 565 *rank-four cells and* 34 *rank-five cells.*

Proof. Use the degree-276 permutation model of $G = \mathrm{Co}_3$ in GAP's primitive-group library. Its identification with Co_3 is a retained library premise. The execution records its generators literally and checks

$$|G| = 495\,766\,656\,000, \qquad Z(G) = 1, \qquad |\operatorname{Aut}(G)| = |G|.$$

Consequently conjugation by G realizes every automorphism of G; the compatible action required by the retained enumeration theorem is the natural action of G itself. The completed enumeration also checks the exact order of the conjugation image against the order of the computed full automorphism group.

For an accepted prefix $t = (t_1, \ldots, t_m)$, its automorphism stabilizer is $S = C_G(t_1, \ldots, t_m)$ and the group containing the eligible next involutions is $Q = C_G(t_1, \ldots, t_{m-1})$. The retained double-coset lemma selects exactly one S-orbit from every involution conjugacy class in Q. The recursive end-intersection criterion then accepts exactly the extensions with the intersection property. Its premise is the already accepted prefix; it is not applied to arbitrary tuples without that premise.

The completion test uses the normal closure in Q of all the eligible class representatives. This is precisely the subgroup generated by every eligible involution, rather than just the selected representatives. Every later mark lies in that subgroup, so the test discards only prefixes that cannot generate G. The complete retained induction proves that the generating leaves represent each automorphism orbit once.

Every new mark lies outside the subgroup generated by its predecessors. The subgroup order therefore increases by a factor of at least two. Since

$$2^{38} \le |G| < 2^{39},$$

the search through rank 38 covers every possible marking. The completed execution records all leaf tuples and all zero rows through this bound. It uses neither the source counts nor a source-imposed rank-five cutoff to direct or stop the search. Its numbers of generating leaves are the three displayed counts. There are no leaves at any other rank, including ranks one and two.

A separate execution checks all 22 908 recorded tuples. For each tuple it verifies the orders of the marks, their membership in the literal group, generation, every string commutation and the full intersection property for every pair of subsets of the marks. For a tuple $(t_1, \ldots, t_r)$ it computes the ordered type

$$(|t_1 t_2|, \ldots, |t_{r-1} t_r|).$$

Grouping those computed types gives the complete distribution. All 824 printed source cells agree, and every computed nonzero type is listed by the source. Thus the comparison also checks the implicit zero assertion for every unlisted type. The source transcription and its physical pages have been separately inspected. The verification of recorded objects supplements the completeness proof; it does not replace that proof or its completed execution. Finally the retained correspondence between regular polytopes and string C-groups identifies these ordered automorphism orbits with the asserted isomorphism classes of polytopes. □

Attribution and finite premises. This recovers the source classification; it is not a new classification claim. Stabilizer enumeration is the method described by Leemans–Mixer, there attributed to Hartley–Hulpke. Double cosets, normal closure and the end-intersection criterion are explicitly retained in the complete common proofs. Exact GAP group operations, the complete conjugacy and automorphism algorithms, and the named permutation model remain software and library premises. The full literal outputs are retained. There is no claim of Lean verification.

The complete L071 recovery also requires its other finite classifications, compatible-action criterion and explicitly stated qualifications. This component alone adds no whole-original or compression-factor credit.